AI APPLICATION ENGINEER

Engineering Talent Career Fair — Hiring Scope & Selection Criteria

Role Definition: An engineer who combines **AI, software engineering, data engineering and industrial/engineering knowledge** to design, build, integrate, deploy and improve AI applications, AI agents and intelligent automation solutions.

Core Value Chain: Engineering Problem → Data → AI/Agent → Application → Integration → Production → Monitoring → Measurable Value

1. Primary Mission

- Identify high-value engineering and manufacturing problems suitable for AI and automation.
- Build AI applications, copilots, agents and intelligent workflows.
- Integrate AI with ERP, MES, CMMS, SCADA, IoT, databases and APIs.
- Move solutions from proof-of-concept to secure, reliable and scalable production.
- Measure impact through productivity, quality, downtime, energy, cost and decision-making improvements.

2. Scope of Responsibility

Area	Expected Capability
AI / GenAI	LLMs, RAG, prompt engineering, embeddings, structured outputs, evaluation and guardrails.
Agentic AI	AI agents, tool calling, workflow orchestration, memory, multi-agent patterns and human-in-the-loop.
Software	Python, REST API/FastAPI, web applications, Git, testing, CI/CD and Docker.
Data	SQL, ETL/ELT, APIs, data pipelines, data quality and structured/unstructured data.
Industrial	IoT, PLC/SCADA, MES, ERP, CMMS, WMS and manufacturing-system integration.
Production	Cloud/on-prem deployment, monitoring, logging, security, reliability and scalability.
Business Value	Translate technical solutions into measurable KPI, productivity, cost and ROI outcomes.

3. Technical Hiring Criteria

Must-Have

- Python or strong software engineering fundamentals
- SQL / database fundamentals
- REST API / application development
- Understanding of Generative AI / LLM concepts
- Git/GitHub and modern development practices
- Ability to build an end-to-end working application
- Strong analytical, problem-solving and learning ability

Strong Advantage

- Agentic AI / LangGraph or equivalent; RAG / GraphRAG / Knowledge Graph
- FastAPI, React, Docker/Kubernetes, CI/CD
- Azure/AWS or enterprise cloud
- IoT, MQTT, OPC UA, PLC / SCADA integration
- MES / ERP / CMMS / WMS integration
- Computer Vision, predictive analytics, time-series
- Local LLM / Ollama and on-prem AI deployment

4. Ideal Candidate

Look for a **builder** who can understand an engineering problem, determine where AI adds value, design the solution, build the application, integrate systems, deploy it and demonstrate measurable results. Prioritize engineering fundamentals, AI capability, problem-solving and learning agility over familiarity with a specific AI framework.

5. Recommended Candidate Backgrounds

Consider candidates from multiple disciplines—not only Computer Science or AI:

- Computer Science / Software Engineering / AI / Machine Learning
- Data Engineering / Data Science
- Electrical / Electronics / Control / Automation Engineering
- Mechatronics / Robotics
- Industrial / Manufacturing Engineering
- Mechanical / Chemical Engineering with strong programming and digital capability

6. Hiring Assessment Framework

Dimension	Weight	Evaluate
AI & Agentic AI	25%	LLM, RAG, agents, AI application design and evaluation
Software Engineering	25%	Python, APIs, architecture, Git, testing and deployment
Engineering / Industrial	20%	Process understanding, systems thinking and integration
Problem Solving	20%	Convert ambiguous problems into practical solutions
Learning & Ownership	10%	Curiosity, initiative, adaptability and self-learning

7. Career Fair Technical Challenge

Challenge: “Choose one real manufacturing or engineering problem and design an AI application to solve it.”

- Define the problem and current process.
- Identify required data and data sources.
- Explain why AI / GenAI / Agentic AI is appropriate.
- Propose the application architecture and system integration.
- Explain security, reliability and hallucination controls.
- Describe deployment, monitoring and human approval.
- Define measurable KPI and expected ROI.

8. Target AI Application Domains

- Production — OEE, monitoring, scheduling and operator copilots
- Engineering — knowledge assistants, troubleshooting and process optimization
- Maintenance — predictive maintenance, failure analysis and maintenance agents
- Quality — defect analysis, root cause and quality prediction
- Warehouse / Supply Chain — forecasting, inventory intelligence and workflow agents
- Energy / Sustainability — optimization, anomaly detection and energy analytics
- EHS / Procurement / Logistics — compliance, document automation and intelligent workflows
- Management — KPI analysis, decision support and executive AI copilots

9. HR Screening Guidance

- **Prioritize:** candidates who have built working applications, automation, APIs, AI agents or integrated systems.
- **Look for evidence:** GitHub projects, internships, university projects, hackathons, deployed applications and measurable outcomes.
- **Do not over-filter:** lack of a specific AI framework should not automatically reject a strong engineering candidate.
- **Test practical ability:** demonstrated building and problem-solving is more valuable than certificates alone.
- **Assess communication:** candidate should explain technical concepts to both engineers and business stakeholders.

10. Candidate Value Proposition

Build the Intelligent Factory. Work at the intersection of **Engineering + AI + Software + Automation + Data**, solving real industrial problems and building solutions that move from idea to production.

- Work on real manufacturing problems with measurable business impact.
- Build AI Agents and intelligent applications—not only models or prototypes.

- Gain exposure to enterprise systems, industrial automation, data platforms and AI infrastructure.
- Develop into a future AI / Engineering technology leader.

11. Recommended Job Position

Primary Title: AI Application Engineer
Alternative: AI & Digital Engineering Application Engineer
Hiring Focus: AI + Agentic AI + Software Engineering + Data + Industrial Integration